

# Deliverables: Dual-Panel Activity & Protocol Visualizer

## 1. Project Links

**GitHub Repository:**

<https://github.com/subhrajyotisaha/NetViz>

**Live Application (Railway):**

<https://netviz.up.railway.app/>

## 2. Project Overview

NetViz is an AI-assisted network protocol simulator and dual-panel web dashboard designed to bridge the gap between user actions and the underlying application-layer protocols. The application allows users to perform three common activities (Browsing, Mail, and Streaming) and simultaneously visualize the resulting network protocol exchanges (DNS, HTTP, SMTP) in real time.

## 3. Key Features

- Dual-Panel Layout: A split-screen interface where user actions are logged on the left, while real-time protocol sequences are animated on the right.
- High-Speed Laser Animations: Network packets are visualized as sleek, fiber-optic data beams that stretch and collapse between the client and server nodes.
- Sequence Diagram Timeline: An automated, animated vertical timeline that constructs a sequence diagram as the protocols execute, allowing users to step forward, backward, or replay.
- Detailed Protocol Payloads: Realistic and highly detailed packet payloads, including full HTTP headers, DNS query flags and sections, and SMTP STARTTLS handshakes.
- Color-Coded Execution: Traffic is uniquely color-coded by protocol (DNS is Blue, HTTP is Yellow, SMTP is Green) to allow for instant visual categorization.

## 4. Technology Stack

- Frontend Structure: Pure HTML5 for semantic structure.
- Styling & Animations: Tailwind CSS via CDN for modern utility classes, combined with custom CSS keyframes for complex laser/timeline animations.
- Interaction & Logic: Vanilla JavaScript (ES6+), utilizing event-driven state machines to govern the synchronization between the activity controls and the visual timeline.
- AI Platform Used: Built primarily via Google Antigravity with Gemini 3.1 Pro (High) for rapid architecture planning, CSS animation generation, and protocol flow mockups.

## 5. Evaluated Protocol Flows

- Web Browsing: Traces a DNS A-record lookup followed by a robust HTTP GET request and standard 200 OK response, demonstrating stateless fetching.
- Email (SMTP): Traces a DNS MX-record lookup, followed by a stateful SMTP conversation including EHLO, STARTTLS encryption, envelope setup, and payload transmission with DKIM/SPF headers.
- Media Streaming: Traces an HTTP-based HLS streaming flow, demonstrating the retrieval of a Master

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Manifest (.m3u8), a Media Playlist, and sequential video data segments (.ts) represented via hex dumps.